

PROTEINLOOP

AMD DEVELOPER HACKATHON · ACT II

Local AI for living protein systems.

Keep fish, prawns, plants, duckweed, and eggs recoverable when cloud access disappears.

GEMMA 4 E2B

local planning

DECT NR+

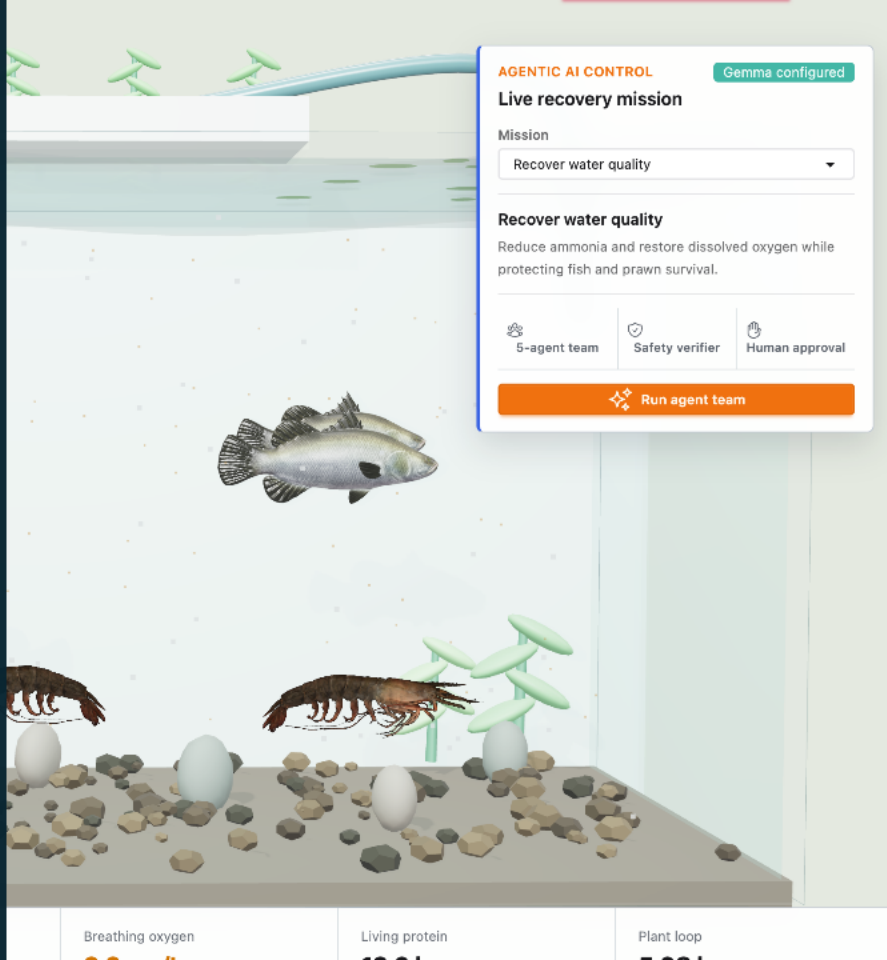
private field link

RULES

deterministic safety

BEST UNICORN

proteinloop.dev-vb.lat



ONE LOOP - SHARED FAILURE

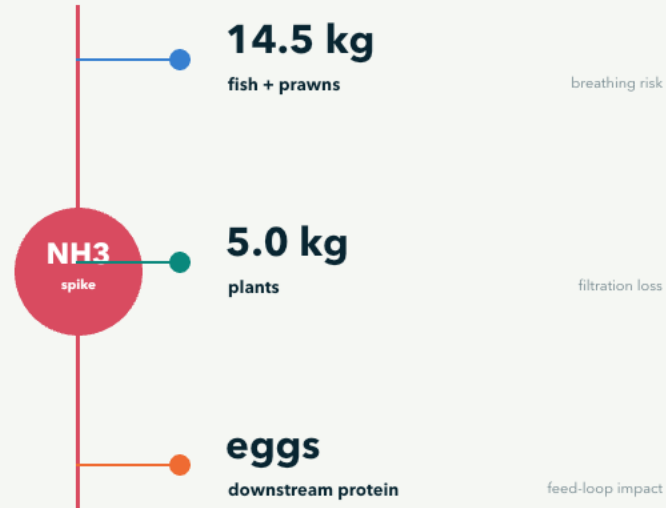
One ammonia spike threatens every protein output.

Aquaponics is usually framed as vegetables. ProteinLoop measures what the same water failure means for animals and downstream food.

ANIMAL PROTEIN AT RISK

14.5 kg

fish + prawn stock breathing the same water



DEPLOYED OPERATOR VIEW

A living system people can understand in seconds.

The tank, chemistry, biomass, and recovery controls share one visual language.

01

ANIMALS

Movement changes with water stress.

02

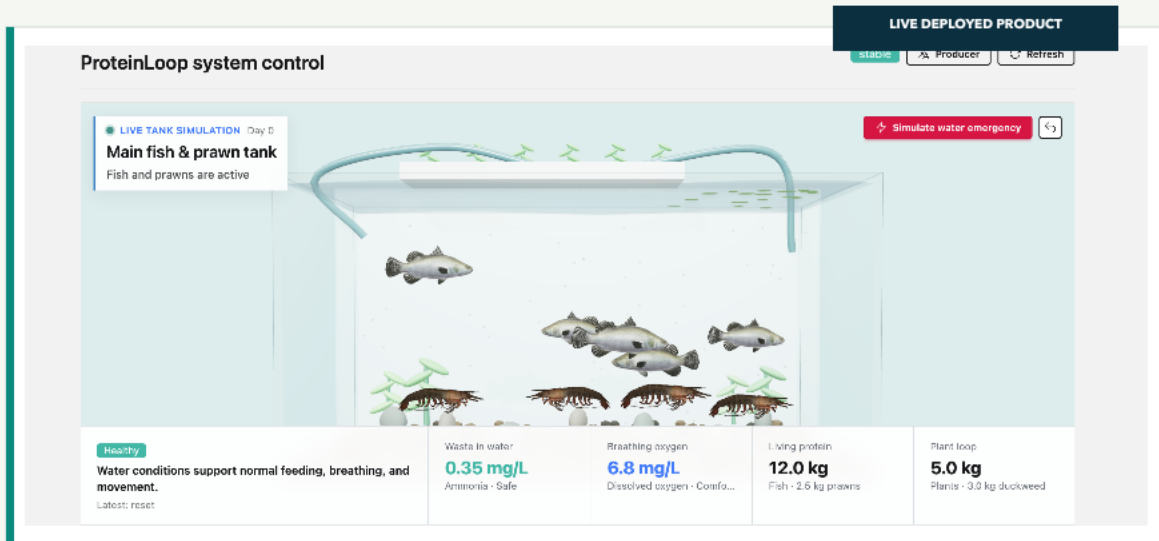
WATER

Ammonia is waste; oxygen is breathable air.

03

ACTION

Verified recovery stays in the same scene.



The recovery is visible while it happens.

Every specialist, verifier outcome, and producer boundary appears as a structured event.

AGENTIC AI CONTROL

Live recovery mission

Mission

Recover water quality

Recover water quality

Reduce ammonia and restore dissolved oxygen while protecting fish and prawn survival.



5-agent team



Safety verifier



Human approval



Run agent team

1

Observe

2

Specialists

3

Supervisor

4

Verify

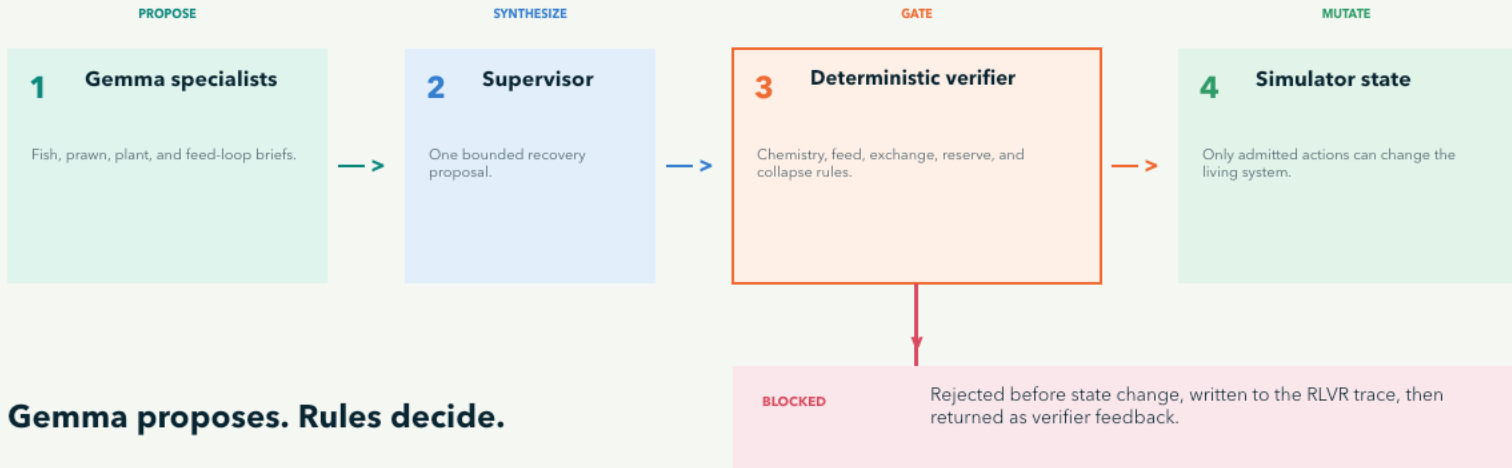
5

Measure

● AUTHORITY, NOT AUTONOMY

Gemma can recommend. It cannot mutate.

Only the deterministic verifier can admit an action into ecosystem state.



Gemma proposes. Rules decide.

● UNICORN MARKET WEDGE

Aquaculture is a \$371B market. The wedge is resilience.

FAO reports 103M tonnes of farmed aquatic animals in 2024. ProteinLoop starts where cloud-first control is a poor fit.

\$371B

2024 FARM-GATE VALUE

103M t

aquaculture aquatic-animal production

WHO PAYS

- Fish + prawn farms
- Producer cooperatives
- Food-security programs

The buyer is the operator accountable for biomass survival and production continuity.

HOW IT EARNS

DEPLOY Edge commissioning + radio gateway

SUBSCRIBE Annual site software + support

EXPAND Optional fleet and evidence service

LAND AND EXPAND

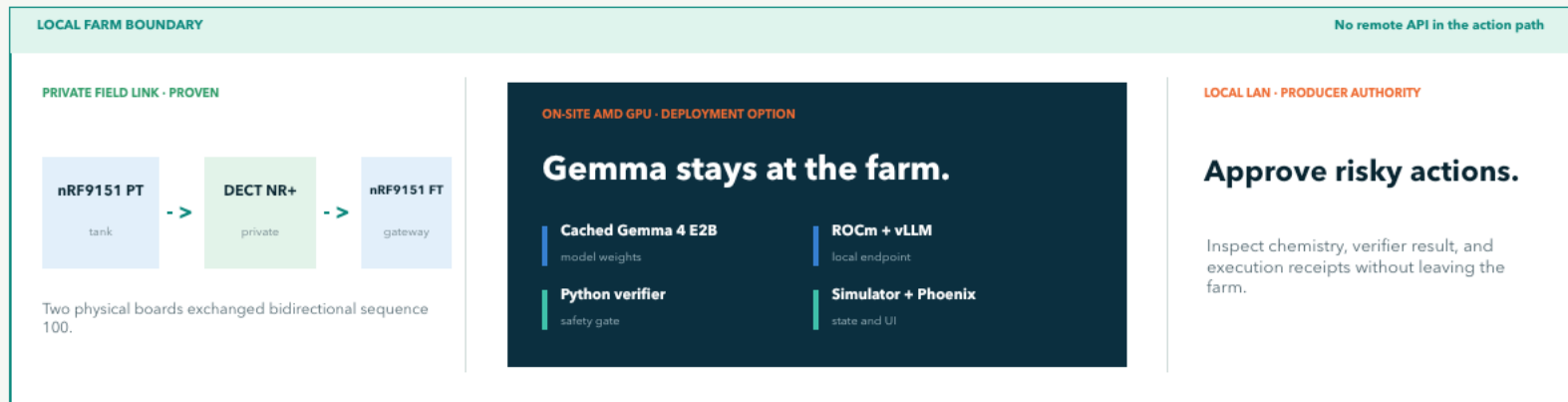
TANK > SITE > COOPERATIVE NETWORK

Revenue grows with protected production capacity, not user screen time.

LOCAL BY DESIGN

An on-site AMD GPU can keep the decision loop local.

After provisioning, cached Gemma inference and deterministic decisions do not require an internet API.

**POWER - ALWAYS REQUIRED**

AMD compute needs electricity; solar + battery autonomy is the next measured field proof.

CLOUD - OPTIONAL

Sync evidence and model updates when connectivity returns; never required to verify or execute.

● HUMAN IN THE LOOP

The workflow pauses at the only irreversible boundary.

The producer sees current chemistry, the proposed action, and the verifier result before anything changes.

LIVE CONTEXT

2.75 mg/L NH3

9.2 mg/L O2

VERIFIED PLAN

Increase aeration. Reduce feed. Monitor the next reading.

PRODUCER DECISION REQUIRED

Protect the protein loop

APPROVE	Execute the verified proposal exactly once.
APPLY HALF	Reduce the amount, then verify the edit again.
REJECT	Resume with zero simulator mutation.

Every choice becomes a receipt: mutation status, chemistry, and verifier reward.

CAPTURED INFERENCE-TIME EVIDENCE

Verifier feedback turns 10% first-pass safety into 100% model-safe plans.

Exact rule failures return as bounded feedback; each revision is parsed and verified again.

FIRST PASS

10%

2 / 20 safe

VERIFIER FEEDBACK

18 unsafe answers repaired

17 in one revision · 1 in two revisions

MODEL SAFE

100%

20 / 20 · zero fallback

MODEL

Gemma 4 E2B

ROCm 7.2 · vLLM 0.20.2 · gfx1100 · 47.98 GiB

EXECUTION

139 requests

60,385 observed tokens · 99.793 tok/s · p50 655.522 ms

COMPLETENESS

363 executable checks

222 Python + 141 Phoenix · zero failures · no weight update

420.648 kg aggregate scenario biomass · +221.7244 mean reward vs naive

From one tank to the operating layer for resilient aquatic food.

Start with hard-to-connect fish and prawn farms. Expand site by site through cooperatives and food-security programs.

BEACHHEAD

Off-grid fish + prawn farms

REVENUE

Deployment + annual software/support

SCALE

Tank > site > cooperative network

ASK · BEST UNICORN + BEST AMD-HOSTED GEMMA

proteinloop.dev-vb.lat · github.com/Anarpego/ProteinLoop

